

Project ID:

R26-SE-007

1. Topic (12 words max)

AI Driven Privacy Preserving Digital Ecosystem for Modernizing Sri Lanka's Paddy Supply Chain

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Sri Lanka's paddy cultivation system historically regarded as the backbone of national food security, currently faces interconnected challenges across production, storage, pricing, and market coordination.[1][2][3][4] Small and medium scale farmers, mill owners, and institutions such as the Sri Lanka Paddy Marketing Board(PMB) are affected by climate variability, fragmented decision making, post harvest losses, price volatility and the lack of trustworthy real time data.[1][2][3] These issues threaten farmer livelihoods reduce national rice self sufficiency and weaken public trust in institutional interventions.[5]

The gravity of the problem is evident at multiple levels. Farmers lack timely, localized guidance on yield optimization, pest management and climate adaptation leading to inefficient input usage and unstable incomes.[6] Institutions such as the (PMB) struggle with transparent warehouse monitoring, disaster time coordination and price stabilization due to reliance on manual and informal reporting systems.[5] Economically, this results in production inefficiencies, avoidable losses and unfair market power imbalances. Socially and ethically, opaque pricing and information asymmetry disadvantage farmers, while environmentally, poor forecasting and reactive practices increase resource waste and climate vulnerability.[1]

This research addresses these challenges through a unified, technology driven approach aligned with the SST Cluster, particularly Software Engineering, Algorithms Design, High Performance and Parallel Computing, Mobile Computing, and Rich Web Based Architectures. The work spans the Agri food systems and digital supply chain industry verticals, integrating AI, distributed systems and secure architectures.[7]

The proposed solution directly advances SDG 2 (Zero Hunger) by improving sustainable paddy production and farmer decision support, SDG 12 (Responsible Consumption and Production) by reducing post harvest losses and enhancing transparency, SDG 13 (Climate Action) through climate aware forecasting and disaster responsive coordination, and SDG 9 (Industry, Innovation, and Infrastructure) via secure, AI enabled modernization of Sri Lanka's rice ecosystem. Collectively, the research contributes a resilient, future ready digital foundation for Sri Lanka's paddy sector. [9] [10]

[1] Paddy price forecasting in India using ARIMA model

https://www.cwss.in/Journal/Complete_journal/Vol.17-No.1_6.pdf

[2] Forecasting of Paddy Grain and Rice's Price: An ARIMA

https://www.academia.edu/86405305/Forecasting_of_Paddy_Grain_and_Rices_Price_An_ARIMA_Autoregressive_Integrated_Moving_Average_Model_Application

[3] Paddy Price Prediction in the South-Western Region of Bangladesh

https://link.springer.com/chapter/10.1007/978-3-030-93247-3_26

[4] Data Analytics in Farming: Rice price prediction in Andhra Pradesh

<https://ieeexplore.ieee.org/abstract/document/10029009>

[5] Research on the Construction of Grain Food Multi-Chain Blockchain Based on Zero-Knowledge Proof (2023) Focuses on blockchain with ZKP for privacy in grain/food supply chains, including warehouse-related data management.

<https://pmc.ncbi.nlm.nih.gov/articles/PMC10138098/>

[6] Crop yield prediction using deep neural networks, Demonstrated that deep neural networks (particularly CNNs and LSTMs) outperform traditional machine learning methods for soybean yield prediction, Khaki, S., & Wang, L. Published: 2019, Frontiers in Plant Science

<https://doi.org/10.3389/fpls.2019.00621>

[7] Zero-Knowledge Proofs in Blockchain-Enabled Supply Chain Management (2024)

Explores ZKP applications in supply chains for privacy-preserving verification, relevant to capacity and stock claims without full disclosure.

https://link.springer.com/chapter/10.1007/978-981-97-0088-2_3

[8] Agriculture-Food Supply Chain Management Based on Blockchain and IoT: A Narrative on Enterprise Blockchain Interoperability (2022) Covers blockchain and IoT integration for warehouse inventory and supply chain management in agriculture.

<https://www.mdpi.com/2077-0472/12/1/40>

[9] Beyond Traceability: Decentralised Identity and Digital Twins for Verifiable Product Identity in Agri-Food Supply Chains (2025) Discusses Decentralized Identifiers (DID) in agricultural supply chains for secure attribution and traceability.

<https://www.mdpi.com/2076-3417/15/11/6062>

[10] Electronic National Agriculture Market (eNAM) and Market Reforms in India, This paper analyzes India's eNAM platform, which digitizes agricultural trade by providing price visibility and online bidding. Chand, R. Published: 2016, *Economic & Political Weekly*
<https://www.mdpi.com/1911-8074/16/2/62>

[11] Deep learning in agriculture: A survey, Comprehensive survey of deep learning applications in agriculture including crop management, livestock management, water management, and soil management. Kamilaris, A., & Prenafeta-Boldú, F.X. published: 2018, *Computers and Electronics in Agriculture* (Elsevier) <https://doi.org/10.1016/j.compag.2018.02.016>

[12] Digital Twin: Enabling Technologies, Challenges and Open Research, Provided comprehensive framework for Digital Twin implementation covering data integration, real-time synchronization, and predictive analytics, Jones, D., et al. Published: 2020, *IEEE Access*
<https://doi.org/10.1109/ACCESS.2020.2998358>

[13] Retrieval-Augmented Generation (RAG) for Knowledge-Intensive NLP Tasks, Introduced RAG architecture that combines parametric memory (neural networks) with non-parametric memory (document retrieval). Lewis, P., et al. (Facebook AI Research & UCL) Published: 2020, *NeurIPS (Neural Information Processing Systems)* <https://arxiv.org/abs/2005.11401>

[14] Secure Matching and Privacy-Preserving Market Mechanisms, Differential Privacy: A Survey of Results, This foundational work introduces Differential Privacy as a mathematical framework to protect individual data in analytical systems, Dwork, C. Published: 2006, *ICALP*
https://dl.acm.org/doi/10.1007/11681878_14

[15] Multi-Agent Systems in Market Modeling, Agent-Based Computational Economics: A Constructive Approach to Economic Theory, This work demonstrates how agent-based models can simulate complex economic systems involving buyers, sellers, and regulators. Tesfatsion, L. Published: 2006, *Handbook of Computational Economics*
[https://doi.org/10.1016/S1574-0021\(05\)02016-2](https://doi.org/10.1016/S1574-0021(05)02016-2)

[16] Recommendation Systems in Agriculture, Big Data and Smart Farming – A Review, The paper reviews data-driven decision support systems in agriculture, emphasizing crop planning, input optimization, and farm management. Wolfert, S., et al. Published: 2017, *Agricultural Systems* (Elsevier)
<https://doi.org/10.1016/j.agsy.2017.01.023>

[17] Graph Neural Networks: A Review of Methods and Applications, Comprehensive review of GNN architectures including Graph Convolutional Networks (GCN), GraphSAGE, and Graph Attention Networks, Zhou, J., et al. Published: 2020, *AI Open* (Elsevier)
<https://doi.org/10.1016/j.aiopen.2021.01.001>

6. Brief description of Existing Research and Systems (300 words max)

Component 1 - Digital Twin for Paddy Farming

Existing agricultural decision support systems mainly provide rule based recommendations or isolated yield predictions using classical machine learning models such as Linear Regression, Random Forest and LSTM. Research has shown improvements using deep learning and computer vision for yield prediction and disease detection. However, current systems lack real time Digital Twin representations, multimodal data fusion and contextual knowledge integration. There is no Sri Lanka specific solution combining Digital Twins, Vision Transformers, RAG based advisory systems and Graph Neural Networks to model inter farm interactions.

Component 2 - AI Driven Paddy Price Forecasting

Current price forecasting tools rely on statistical models (ARIMA, SARIMA) or basic ML models focused only on short term prediction accuracy. Prior research rarely addresses interpretability or causal reasoning behind price changes. Existing systems do not integrate government policies, climate effects or market behavior simulation. There is no unified framework for long range forecasting, causal explanation and multi agent simulation of the Sri Lankan rice market.

Component 3 - Secure, Disaster Aware Warehouse Coordination

Warehouse management in Sri Lanka primarily depends on manual reporting, spreadsheets, and informal communication channels. While blockchain and IoT have been studied separately in logistics research, practical agricultural implementations remain limited. Existing systems lack verifiable stock records, privacy preserving data sharing and disaster time coordination mechanisms. No prior work integrates blockchain, decentralized identity, zero-knowledge proofs and AI-based reliability scoring for paddy warehouse networks.

Component 4 - AI Powered Farmer Market Marketplace

Existing digital marketplaces provide basic price visibility but lack intelligence, privacy and negotiation support. Research on federated learning, secure matching and AI negotiation exists in other domains but is rarely applied to agriculture. Current systems do not protect farmer data, support automated negotiation or offer personalized market guidance using LLMs. There is no Sri Lanka focused, privacy preserving AI marketplace tailored to paddy supply chains.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

The proposed research introduces a unified, AI driven, privacy preserving digital ecosystem to address the critical gaps identified in Section 6 across Sri Lanka's paddy production, pricing, storage and market coordination processes. Unlike existing fragmented or rule based systems. The solution adopts a component integrated system of systems approach, ensuring real time intelligence, explainability, trust and resilience across the entire paddy value chain.

Component 1 directly addresses the absence of real time Digital Twins and multimodal intelligence by creating a continuously updated virtual representation of paddy fields. It integrates climate data, soil metrics, farm metadata and field images to overcome the limitations of isolated yield prediction models. The use of Vision Transformers improves crop health analysis, while Graph Neural Networks model inter farm dependencies such as disease spread and water flow. A RAG based advisory system bridges the knowledge gap by grounding recommendations in Sri Lankan agricultural guidelines, overcoming the lack of contextual, explainable farmer support.

Component 2 resolves the limitations of short term and opaque price forecasting systems by introducing long range Transformer based models combined with causal machine learning. This allows not only accurate price predictions but also explanations of why prices change. The inclusion of multi agent simulation fills the gap in understanding market behavior by modeling interactions between farmers, millers, traders and government actors. enabling policy aware and climate sensitive economic analysis specific to Sri Lanka.

Component 3 addresses the lack of trust, transparency and disaster-time coordination in warehouse management by integrating blockchain based immutable records, decentralized identities, and zero-knowledge proofs. This combination overcomes the shortcomings of manual reporting and insecure data sharing, while AI based reliability scoring and lightweight IoT support ensure verifiable and accountable decision making during emergencies.

Component 4 fills the intelligence and privacy gaps in existing digital marketplaces by introducing federated learning, secure matching and AI driven negotiation. This ensures farmer data remains private while enabling personalized recommendations and fair, automated farmer miller negotiations capabilities absent in current agricultural platforms.

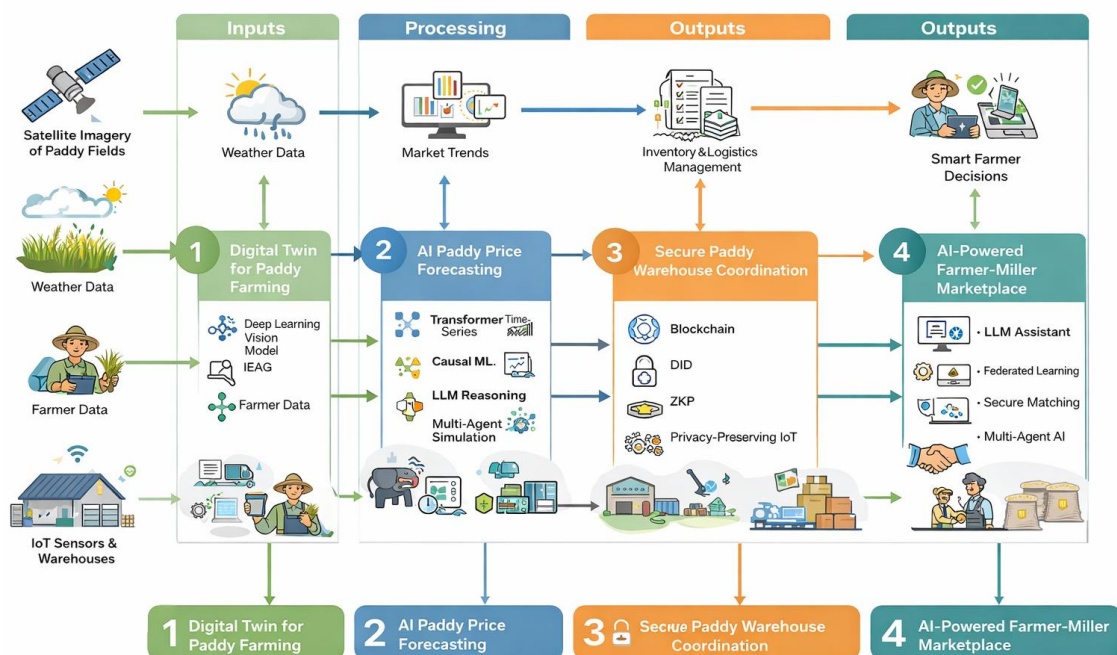
Inputs - Multimodal farm data(images, soil, weather), historical price data, climate indicators, government policy documents, warehouse stock and sensor data and marketplace transaction metadata.

Processes - AI driven data fusion, Digital Twin simulation, Transformer based forecasting, causal reasoning, multi agent market simulation, blockchain secured record management, privacy preserving cryptographic verification, federated learning and LLM based RAG assistance.

Outputs - Actionable farmer advisories, real time yield and risk predictions, explainable price forecasts, verified warehouse coordination decisions, disaster responsive redistribution plans and a secure intelligent farmer miller marketplace with automated negotiation support.

Overall, the solution transforms Sri Lanka's paddy ecosystem into a data driven, transparent and resilient digital infrastructure, addressing long standing technical, economic and institutional limitations while enabling sustainable and future ready agricultural development.

AI-Driven, Privacy-Preserving Digital Ecosystem for Modernizing Sri Lanka's Paddy Supply Chain



8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The successful implementation of the proposed research depends on the availability of agricultural domain expertise, advanced technical knowledge and reliable data sources. The project team has identified clear and feasible pathways to acquire and integrate these resources throughout the research lifecycle.

Specialized domain knowledge related to paddy cultivation, rice supply chains and institutional operations will be obtained through direct collaboration with the Sri Lanka Paddy Marketing Board(PMB). The research problem itself was identified with the guidance of a (PMB) representative to ensuring that the project addresses real operational challenges. Domain expertise will be strengthened by consulting agricultural research publications, policy documents and datasets provided by the Hector Kobbekaduwa Agrarian Research and Training Institute(HARTI). These sources offer authoritative insights into crop production trends, how prices fluctuate, post harvest practices and policy driven market interventions specific to Sri Lanka.

Where direct field access is limited, expert knowledge will be supplemented through structured interviews, publicly available reports and documented best practices from government and research institutions. This multi source approach ensures both practical relevance and academic rigor.

The technical competencies required such as artificial intelligence, deep learning, time-series forecasting, blockchain systems, federated learning and distributed architectures will be acquired through a combination of formal academic learning and self directed study. Team members will leverage online platforms, including research journals, official framework documentation, open source repositories, technical blogs and curated YouTube lecture series to gain hands on experience with relevant tools and technologies.

Existing open source projects and prior academic implementations will be analyzed and adapted to accelerate skill acquisition. Continuous consultation with academic research supervisors and iterative prototyping will further strengthen technical proficiency and system design expertise.

The project primarily relies on secondary and institutional data sources, minimizing feasibility risks. Required data including historical paddy prices, production statistics, climate data, warehouse records and policy documents will be collected from PMB records, HARTI databases, government open data portals and publicly available meteorological sources. Image and simulated farm data may be obtained from open agricultural datasets or generated through controlled experimental setups.

Where real time data access is constrained, synthetic or anonymized datasets will be used for model training and validation, ensuring robustness without compromising realism.

The project does not involve human experimentation or the collection of personal sensitive data. However, ethical considerations arise in data privacy, institutional data usage and AI driven decision support. Any institutional data obtained from PMB or HARTI will be used strictly with authorization, anonymized where necessary and handled in compliance with academic data governance standards. If required by institutional guidelines and formal ethical clearance will be sought prior to data acquisition.

Overall, the project benefits from strong institutional collaboration, accessible technical learning resources and ethically manageable data requirements to ensuring its feasibility and research integrity.

9. Objectives and Novelty

Main Objective To design and develop a secure, AI driven and privacy preserving digital ecosystem that integrates intelligent farming support, explainable price forecasting, trusted warehouse coordination and a fair farmer miller marketplace in order to enhance productivity, transparency, climate resilience and decision making across Sri Lanka's paddy cultivation and supply chain.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22102096 Ransara N.S.	- To develop a long range paddy price forecasting model using Transformer based time series architectures tailored to Sri Lanka. - To identify and explain the causal factors influencing paddy price fluctuations, including climate, policy, and supply chain dynamics.	- Collect and preprocess historical paddy price data, climate data and policy related datasets relevant to Sri Lanka. - Design and implement Transformer based forecasting models(Informer/FEDformer) and benchmark them against ARIMA and LSTM models. - Apply causal machine learning techniques to identify key drivers behind	- First Sri Lanka specific paddy price forecasting system combining Transformer based long range prediction, causal machine learning and multi agent market simulation. - Moves beyond prediction only systems by explaining why prices change using counterfactual and causal reasoning. - Introduces agent based simulation to analyze policy and climate scenarios before real world implementation.

	To simulate the behavior of key market stakeholders using a multi agent framework for policy and scenario analysis.	price changes and generate interpretable explanations. - Develop a multi agent simulation modeling farmers, millers, traders and government entities. - Integrate forecasting, causal reasoning and simulation outputs into a unified analytical framework. - Evaluate model performance, interpretability and scenario outcomes using real and simulated data.	Provides an integrated, explainable AI framework rather than isolated forecasting models.
IT22294784 Kumarasingha P.A.N.D	Development of an intelligent Digital Twin system for paddy farming that integrates multimodal data analysis, knowledge-augmented advisory services, and inter-farm relationship modeling to enable real-time yield prediction, crop health monitoring, and data-driven decision support for farmers.	- Gather multimodal dataset including field imagery, soil data, weather parameters, and yield records. - Preprocess data ensuring consistency and temporal alignment. - Develop Temporal Fusion Transformer for yield prediction comparing against LSTM and XGBoost. - Design Vision Transformer for crop	This Digital Twin uniquely combines Temporal Fusion Transformers for yield forecasting, Vision Transformers for image analysis, and RAG for knowledge-grounded recommendations. Graph Neural Networks model farm-to-farm interactions—water sharing, pest migration, disease spread—addressing critical gaps. Unlike existing systems with isolated data, this framework delivers multimodal integration, real-time adaptation, superior prediction accuracy, and holistic network understanding, creating an unprecedented platform for Sri Lanka's paddy farming.

		health, pest, and disease detection. • Build RAG system integrating agricultural manuals and fertilizer guidelines. • Construct GNN models for inter-farm water networks and pest propagation. • Validate system accuracy across agro-ecological zones. • Develop farmer interface with local language support.	
IT22147950 Chamudi K. S. I.	To design and implement an AI-powered, privacy-preserving digital marketplace that enables fair and intelligent price negotiation between farmers and millers in the paddy supply chain.	• Analyze limitations of existing farmer-miller digital marketplaces • Design a privacy-preserving data sharing mechanism using Federated Learning • Develop a price prediction model based on historical paddy price data • Integrate a RAG-based LLM assistant to provide market insights and selling guidance • Implement a secure farmer-miller matching mechanism based on price, quantity, and location • Design and simulate an LLM-based autonomous negotiation framework • Evaluate system performance, fairness, and privacy effectiveness	• First application of Federated Learning in a farmer-miller marketplace to protect sensitive agricultural data • Introduction of a RAG-based intelligent assistant that provides explainable and evidence-based market guidance instead of generic chatbot responses • Novel identity-preserving matching mechanism that prevents early exposure and price exploitation • Use of LLM-based autonomous agents to conduct real-time, goal-oriented price negotiations • Unified AI marketplace architecture combining privacy preservation, intelligent reasoning, and automated negotiation for paddy trading

IT22372444 Senarathne S.M.B.V.B.	To design and implement a secure, disaster aware paddy warehouse coordination system using blockchain for immutable records, decentralized identity for accountability, lightweight IoT for data integrity, zero knowledge proofs for privacy preserving capacity verification for anomaly detection and reliability scoring, enabling trusted and rapid redistribution decisions during emergencies in Sri Lanka's regional warehouse network.	○ Design and implement a secure, disaster-aware warehouse coordination system integrating blockchain, decentralized identity (DID), zero knowledge proofs (ZKP), lightweight IoT sensors for AI based anomaly detection. ○ Develop a blockchain-based immutable ledger for recording warehouse events (inbound/outbound stock, redistribution, disaster damages) with timestamps, digital signatures, and off chain data hashes. ○ Implement decentralized identity (DID) for warehouses and Regional Managers to enable secure authentication, cryptographic signing of reports, and non-repudiable attribution. ○ Integrate zero knowledge proofs (ZKP) for privacy preserving verification of capacity claims (e.g., proving "capacity $\geq$ X tons" without revealing exact values).	○ First integration of a disaster-aware blockchain framework tailored for agricultural warehousing in paddy supply chains. ○ Novel application of zero knowledge proofs (ZKP) for privacy preserving capacity and loss verification in warehouse coordination. ○ Introduction of lightweight IoT integration with blockchain anchoring for low complexity data integrity support in resource constrained environments. ○ A context specific solution for Sri Lanka's regional paddy storage system, addressing real challenges like informal coordination, elephant attacks, floods, and post disaster audits. Globally, blockchain/ZKP/DID/GNN are applied to agricultural supply chains, but no Sri Lankan specific studies combine them for warehouse management, especially in disaster contexts
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		○ Deploy lightweight IoT sensors (environmental, stock trend indicators) with off-chain storage and on chain hashing for independent data integrity and anomaly support. ○ Design and simulate the disaster response workflow, including event recording, capacity proof submission, ranking of warehouses (by distance, capacity, reliability), and signed redistribution orders.	
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**10. This part is to be filled by the
Supervisor and the Co-supervisor of the Project.**

- a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	☒	No	☐
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- b) The proposed topic exhibits novelty.

Yes	☒	No	☐
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- c) The student group can successfully execute the proposed project.

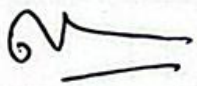
Yes	☒	No	☐
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- d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	☒	No	☐
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- e) Any other comments:

Proceed.

	Title	First Name	Last Name	Signature
Supervisor	Dr.	Mahima	Weerasinghe	
Co-Supervisor	Mr.	Eishan	Weerasinghe	
External Supervisor	Mr.	Suranga	Senanayake	
Summary of external supervisor's (if any) experience and expertise Regional Manager, Paddy Marketing Board MABM (University of Ruhuna), B.Sc. Agric. Specializing in Food Science (Wayamba University), DHRM (University of Colombo)				